PART X

Practical Project Progression

Keep evolving **one realistic production project** as you learn each technology. This turns individual tools into a coherent DevOps skill set.

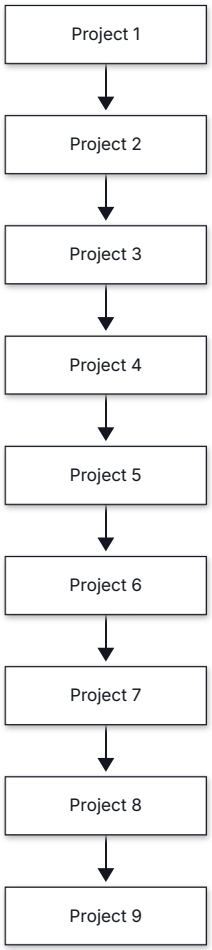


Figure 1

#	Project	What you build	Levels used
1	Project 1 - Basic Server	Ubuntu + Nginx + Node.js + PostgreSQL deployed manually.	Levels 0-2

#	Project	What you build	Levels used
2	Project 2 - Docker	Node.js + PostgreSQL + Redis + Nginx containerized with Docker Compose.	Level 4
3	Project 3 - CI/CD	GitHub → GitHub Actions → tests → Docker → registry → deployment.	Levels 3, 5
4	Project 4 - Terraform	Provision AWS VPC, subnets, EC2, ALB and RDS using Terraform.	Levels 6-7
5	Project 5 - Kubernetes	Move the application to Kubernetes.	Level 8
6	Project 6 - Production Kubernetes	Add Ingress, TLS, HPA, RBAC, Secrets, NetworkPolicies, Helm and Argo CD.	Levels 8, 9, 16, 17
7	Project 7 - Observability	Add Prometheus, Grafana, Loki, OpenTelemetry and Sentry.	Level 10
8	Project 8 - Disaster Recovery	Simulate infrastructure and database failures and recover from them.	Levels 13-15
9	Project 9 - Senior Architecture	Build a complete production-style platform using AWS, Terraform, Docker, Kubernetes, Helm, GitOps and observability.	Levels 6-20